UNIT–1 MINI PROJECT

Enterprise Student Record and Academic Management Platform

Design Thinking, Product Planning & DevOps-Ready Engineering

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Subject	DevOps – Unit 1 Mini Project

Prepared according to the Unit–1 Mini Project Guide

1. Project Overview

Project Title: Enterprise Student Record and Academic Management Platform (ESRAMP)

The project is a proposed software product for managing student records and academic information in a higher-education institution. The Unit–1 guide requires students to understand the academic domain, identify stakeholders, analyze user needs, define requirements, prepare the product vision and roadmap, and initialize the project for later Agile development. This document consolidates those required deliverables into one submission-ready report.

Important scope note: The uploaded project guide is focused on product discovery and planning rather than full application implementation. Therefore, the architecture, technology choices, repository structure, and roadmap below are proposed engineering plans for the next stages, not claims that the complete platform has already been implemented.

1.1 Project Objective

- Create a centralized platform for student academic records.
- Reduce dependency on manual records and disconnected spreadsheets/files.
- Support student registration, attendance, examination/result management, and academic report generation.
- Provide role-based access for students, faculty, academic administrators, examination staff, and IT support.
- Prepare a clear product foundation for Agile development and later DevOps automation.

1.2 Business Problem Statement

Educational institutions manage large amounts of student information across admission, course registration, attendance, examinations, results, and academic reporting. When these activities are handled through manual records or disconnected systems, information can become difficult to find, duplicate, inconsistent, or slow to update. Students and faculty may also lack a single, reliable view of academic information.

Core Problem Statement: Students, faculty members, and academic administrators need a centralized and reliable way to manage student and academic information because fragmented/manual record management can create delays, duplication, reporting effort, and limited visibility.

2. Business Requirements Document (BRD)

2.1 Business Understanding

The platform focuses on the academic management process described in the mini-project guide: student admissions, course registration, attendance, examination, result management, and reporting.

Business Area	Current/Expected Activity	Platform Need
Student Admission & Registration	Create and maintain student academic records.	Central student profile and registration module.
Course Registration	Assign students to programs/courses/subjects.	Course and enrollment management.
Attendance	Record attendance against subjects and students.	Faculty attendance entry and student view.
Examination	Manage examination-related academic information.	Exam and assessment records.
Result Management	Store and publish marks/results.	Result entry, calculation and publishing workflow.
Academic Reporting	Prepare student and administrative reports.	Searchable reports and export-ready summaries.

2.2 Business Objectives

1. Centralize academic records in one platform.

2. Make important student information easier to access for authorized users.
3. Reduce repeated manual data entry and reporting effort.
4. Improve consistency and traceability of academic records.
5. Create a scalable foundation that can later use CI/CD, containers, infrastructure automation, and monitoring.

2.3 Expected Business Benefits

- Faster retrieval of student academic information.
- Better visibility for faculty and academic administrators.
- Reduced duplication and avoidable data-entry errors.
- More structured academic reporting.
- Clear ownership and access control for different user roles.
- A development process that can evolve from product discovery into Agile and DevOps delivery.

3. Stakeholder Analysis Report

The project guide specifically identifies students, faculty members, academic administrators, examination cell staff, and IT support personnel as stakeholders. The following matrix defines their expected involvement.

Stakeholder	Role	Needs / Expectations	Project Responsibility
Students	Primary users	View profile, attendance, results and academic reports.	Use the system and provide usability feedback.
Faculty Members	Academic users	Manage attendance and academic information for assigned subjects.	Enter/verify academic data and validate workflows.
Academic Administrators	Management users	Maintain student/course records and generate reports.	Approve processes and monitor data.
Examination Cell Staff	Examination operations	Manage examination/result information.	Maintain examination records and result workflows.
IT Support	Operations	Reliable, secure and maintainable platform.	User support, deployment and operational maintenance.
College Management	Business stakeholder	Visibility, compliance and improved academic operations.	Set priorities and evaluate business outcomes.

4. User Persona Report

Personas convert stakeholder needs into realistic user-centered requirements.

Persona	Goals	Responsibilities	Challenges	Expected Features
Aarav – Student	Access academic information quickly.	View personal academic records and reports.	Searching for information across multiple sources.	Student dashboard, attendance, results, reports.
Neha – Faculty Member	Maintain accurate academic records.	Record attendance and academic information.	Repeated manual entry and verification.	Faculty dashboard, attendance entry, subject-wise records.
Dr. Patil – Academic Administrator	Monitor academic operations.	Manage records and generate reports.	Manual report preparation and limited centralized visibility.	Admin dashboard, search, reports and controls.
Ms. Rao – Examination Staff	Manage examination/result data accurately.	Enter and verify examination information.	Data consistency and result-processing workload.	Exam/result module, validation and report generation.

5. Design Thinking Report

The project guide requires five stages: Empathize, Define, Ideate, Prototype and Test. The platform is planned by following these stages before full-scale coding.

5.1 Empathize

The team first studies the academic environment and the experiences of students, faculty members, administrators, examination staff and IT support. The focus is on understanding where information is created, how it is updated, who needs it, and where delays or confusion occur.

- Observe student academic information access needs.
- Understand faculty attendance and academic record workflows.
- Understand administrative and examination reporting needs.
- Identify difficulties caused by manual or fragmented record management.

5.2 Define

Defined problem: The institution needs a centralized academic record platform because fragmented/manual handling of student, attendance, examination and result information can increase effort, delay access and make reporting difficult.

5.3 Ideate

Potential solution ideas generated from the defined problem:

- Central student profile and academic record.
- Role-based dashboards for students, faculty and administrators.
- Attendance management with subject-wise views.
- Examination and result management.
- Academic report generation.
- Search and filtering for authorized users.
- Notifications for important academic updates as a future enhancement.

5.4 Prototype

A low-fidelity prototype should be prepared before implementation. The first prototype can contain the following screens:

6. Login / role selection screen.
7. Student dashboard.
8. Faculty attendance dashboard.
9. Administrator dashboard.
10. Student record/search screen.
11. Examination and result screen.
12. Academic report screen.

5.5 Test

The prototype should be demonstrated to representative users. Testing should focus on navigation clarity, ease of finding records, attendance entry, result viewing, report generation and whether the workflow matches the users' actual needs. Feedback should be recorded and used for another iteration.

6. IBM Design Thinking Report

The project guide requires three IBM Design Thinking principles: Focus on User Outcomes, Restless Reinvention, and Diverse Empowered Teams.

6.1 Focus on User Outcomes

The project should measure success through user outcomes rather than only technical completion. Example: instead of only saying 'build a student database search', the outcome is 'enable an authorized user to find a student's required academic information quickly through a simple workflow.'

6.2 Restless Reinvention

The platform should be treated as an evolving product. After each development/release cycle, user feedback, usage observations, defects and performance information should be reviewed. Improvements should then be planned for the next iteration.

6.3 Diverse Empowered Teams

A cross-functional team should include product/academic representatives, UI/UX contributors, software developers, QA/testers, database/infrastructure contributors and DevOps/IT support. This reduces communication gaps and allows decisions to consider both user needs and engineering constraints.

6.4 IBM Loop Documentation – Observe → Reflect → Make

Loop Stage	Application to ESRAMP
Observe	Study user interactions, academic workflows, feedback, defects and operational difficulties.
Reflect	Analyze feedback, identify priorities, compare outcomes with requirements and decide what needs improvement.
Make	Design or modify the product, implement changes, test them and prepare the next release.

7. Functional Requirements Specification (FRS)

The functional requirements below are derived from the project guide's examples of student registration, attendance management, result generation and report generation, and are organized into a practical MVP.

ID	Functional Requirement	Description	Priority
FR-01	User Authentication	Authorized users can sign in according to their assigned role.	High
FR-02	Student Registration	Create and maintain student registration records.	High
FR-03	Student Profile	Store and display authorized student academic information.	High
FR-04	Course Registration	Maintain student course/subject enrollment information.	High
FR-05	Attendance Management	Faculty can record and view attendance for assigned subjects.	High
FR-06	Examination Management	Maintain examination and assessment information.	Medium
FR-07	Result Management	Enter, verify and display student results.	High
FR-08	Report Generation	Generate student and administrative academic reports.	High
FR-09	Search & Filter	Authorized users can search relevant records efficiently.	Medium
FR-10	Role-Based Access	Limit features and records according to user role.	High

8. Non-Functional Requirements Specification (NFRS)

ID	Requirement	Proposed Target / Expectation
NFR-01	Security	Role-based access, secure authentication, protected sensitive academic data and controlled permissions.
NFR-02	Usability	Simple navigation and clear role-specific dashboards suitable for students and staff.
NFR-03	Reliability	Consistent record storage and validation with suitable backup/recovery planning.
NFR-04	Performance	Common searches and dashboard actions should respond quickly under expected academic workload.
NFR-05	Scalability	Architecture should allow additional users, courses and academic records without redesigning the whole system.
NFR-06	Maintainability	Modular code, documented repository structure, tests and clear separation of responsibilities.
NFR-07	Availability	Deployment should be designed for dependable access during normal academic

		operations.
NFR-08	Auditability	Important academic changes should be traceable through appropriate logs/history.

9. Product Vision Document

9.1 Vision Statement

To provide a centralized, secure and user-friendly academic management platform that makes student records, attendance, examinations, results and academic reports easier to manage and access across a higher-education institution.

9.2 Mission Statement

To simplify academic record management by connecting students, faculty and academic administrators through reliable digital workflows while creating an engineering foundation that can be developed and operated using Agile and DevOps practices.

9.3 Business Objectives

- Centralize academic information.
- Improve accessibility of authorized student records.
- Reduce manual effort in attendance and reporting workflows.
- Improve consistency and traceability of academic information.
- Provide a foundation for iterative development and automated delivery.

9.4 Success Criteria

- Users can complete core MVP workflows without unnecessary manual steps.
- Authorized users can locate required academic records efficiently.
- Attendance and result records are stored consistently.
- Academic reports can be generated from centralized data.
- The repository and project process are organized for future Agile/DevOps implementation.

10. Minimum Viable Product (MVP) Definition

The MVP is the first practical release containing only the core features necessary to demonstrate the value of the Student Record and Academic Management Platform.

MVP Feature	Included Capability	Primary Users
User Login & Roles	Authentication and role-based access.	All users
Student Registration	Create/update student academic profile.	Admin / authorized staff
Record Management	Store and view core student records.	Admin / faculty / student
Attendance	Record and view attendance.	Faculty / student
Examination & Results	Manage and display result information.	Exam staff / faculty / student
Academic Reports	Generate basic academic reports.	Admin / student / authorized staff

Deferred for later releases: advanced analytics, AI-based recommendations, complex third-party integrations, and other non-essential capabilities. These are intentionally outside the first MVP so that the team can validate the core product first.

11. Product Roadmap

Phase	Focus	Major Deliverables

Phase 1	Domain & Product Discovery	Business understanding, stakeholders, personas, problem statement, BRD.
Phase 2	Design Thinking & Prototyping	Empathy findings, ideation, wireframes, prototype feedback.
Phase 3	Agile Planning	Backlog, user stories, sprint planning, acceptance criteria.
Phase 4	Core Development	Frontend, backend, database and MVP workflows.
Phase 5	DevOps Automation	Git workflow, CI pipeline, automated tests, containerization and CD planning.
Phase 6	Infrastructure Automation	Infrastructure-as-Code and configuration management planning/implementation.
Phase 7	Deployment & Monitoring	Production deployment, logging, monitoring and continuous improvement.

12. Initial System Architecture Diagram

The following is the proposed architecture for later implementation. It is intentionally technology-neutral at Unit–1 level, while showing the major engineering layers.

Users **Student | Faculty | Academic Admin | Examination Staff | IT Support**
Presentation Layer Web / Responsive UI → Role-Based Dashboards
Application Layer Authentication | Student Records | Attendance | Examination | Results | Reports
Data Layer Relational Database | Validation | Audit/History | Backup/Recovery
DevOps & Operations **Git → CI/CD → Testing → Containerization → Deployment → Monitoring**

Data flow concept: Users access role-specific dashboards → application services validate permissions and process academic operations → database stores authorized records → reports and dashboards present required information → DevOps tooling supports testing, delivery, deployment and monitoring.

13. Project Charter

Item	Project Definition
Project Name	Enterprise Student Record and Academic Management Platform
Project Owner / Student	Sarthak Abhale
PRN	2125UDSM1032
Class	SY-AIDS
Purpose	Design a centralized platform for managing student academic information.
Primary Scope	Student registration, records, attendance, examinations, results and reports.
Primary Users	Students, faculty, academic administrators, examination staff and IT support.
Success Direction	Usable MVP, clear requirements, organized repository and DevOps-ready engineering plan.
Constraints	Unit–1 emphasizes discovery and product planning; full implementation is planned for subsequent stages.
Key Risks	Incomplete requirements, access-control mistakes, inconsistent data, scope expansion and insufficient user feedback.
Risk Response	Iterative requirements review, role-based design, validation, testing, user feedback and controlled MVP scope.

13.1 Milestone Plan

Milestone	Outcome
M1	Business requirements and problem statement completed.
M2	Stakeholders and personas documented.
M3	Design Thinking and IBM Design Thinking documentation completed.
M4	FRS/NFRS and product vision completed.
M5	MVP and roadmap finalized.

M6	Initial architecture and repository structure prepared.
M7	Project is ready for Agile development and later DevOps implementation.

14. Project Repository Initialization

A standardized repository makes the project easier to understand, develop, test and operate. The following structure is proposed for this mini project and later development.

```
student-record-academic-platform/
├── .github/
│   └── workflows/              # Future CI/CD workflows
├── app/
│   ├── frontend/               # User interface
│   ├── backend/                # Application/API services
│   ├── modules/                 # Students, attendance, exams, results, reports
│   └── tests/                   # Unit and integration tests
├── database/
│   ├── schema/                  # Database design
│   └── migrations/              # Future schema migrations
├── docs/
│   ├── BRD.md
│   ├── FRS.md
│   ├── NFRS.md
│   ├── personas.md
│   ├── design-thinking.md
│   └── architecture.md
├── docker/                      # Future containerization files
├── infrastructure/              # Future IaC files
├── monitoring/                  # Future monitoring configuration
├── README.md
└── .gitignore
```

14.1 Git Initialization Plan

13. Create the project repository with the standardized folder structure.
14. Add README.md containing project objective, scope and setup information.
15. Create the initial commit for Unit–1 documentation.
16. Use meaningful branches/commits when development begins.
17. Keep application code, documentation and infrastructure configuration organized in separate folders.

15. Product Discovery Documentation

15.1 Discovery Questions

- Who creates and updates student academic records?
- What information does a student need most frequently?
- How is attendance currently recorded and verified?
- How are examination and result records prepared?
- Which reports are required by students, faculty and administrators?
- Which roles should be allowed to view or modify each record?
- What are the most common causes of delay or duplicate data?
- What feedback should be collected before finalizing the prototype?

15.2 Assumptions to Validate

- The institution requires role-based access to academic information.
- Student, faculty and administrator workflows are different.
- Core MVP records can be represented in a structured relational data model.
- User feedback will be available during prototype testing.
- Future units will define detailed technology and deployment choices.

16. Deliverable Traceability Matrix

Guide Deliverable	Where Covered in This Report
Business Requirements Document (BRD)	Section 2
Product Vision Document	Section 9
Stakeholder Analysis Report	Section 3
User Persona Report	Section 4
Functional Requirements Specification (FRS)	Section 7
Non-Functional Requirements Specification (NFRS)	Section 8
Design Thinking Report	Section 5
IBM Design Thinking Report	Section 6
IBM Loop Documentation	Section 6.4
MVP Definition	Section 10
Product Roadmap	Section 11
Initial System Architecture Diagram	Section 12
Project Charter	Section 13
Project Repository Initialization	Section 14
Project Folder Structure	Section 14
Product Discovery Documentation	Section 15

17. Conclusion

The Enterprise Student Record and Academic Management Platform has been planned as a product-engineering project rather than as only a coding task. The report begins with the educational domain and business problem, identifies stakeholders and personas, applies the five Design Thinking stages, documents IBM Design Thinking and the Observe → Reflect → Make loop, defines functional and non-functional requirements, establishes the product vision and MVP, and prepares a phased roadmap and repository structure.

The result is a structured, DevOps-ready foundation for the next development stages. The actual implementation, CI/CD pipeline, containerization, infrastructure automation, deployment and monitoring should be completed in later units according to the course progression.

18. Source Basis

This report is prepared primarily from the uploaded Unit–1 Mini Project Guide, which specifies the objective, student activities and required deliverables for the Enterprise Student Record and Academic Management Platform. The detailed project content above follows that structure and extends it into a coherent submission document where the guide leaves implementation details open.

The uploaded training document also emphasizes software product engineering, Design Thinking, IBM Design Thinking, requirements engineering, MVP planning, roadmapping, architecture and DevOps automation as part of the broader engineering lifecycle.